



Ray Tracing of Light Trapping Strategies in Thin Silicon Solar Cells based on Tunnel Oxide Passivated Contact (TOPCon)

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Abstract

Tunnel oxide passivated contact (TOPCon) silicon (Si) solar cells, which utilise n-type silicon wafers, thin oxide layers, and polycrystalline silicon, are gaining popularity in photovoltaic industry due to its high efficiency potential. While 160 μm Si wafers are often used in current mass production, thinner wafers are projected to become mainstream in the future to reduce production costs. However, the decrease in the Si thickness reduces light absorptance, leading to lower device efficiency. Light-trapping strategies are utilised to enhance light absorptance in thin TOPCon cells and compensate for the efficiency loss. This work investigates light-trapping strategies in 100 μm -thin TOPCon cells using SunSolve. Upright pyramids with various heights and double layer anti-reflective coating (DLARC) silicon dioxide/silicon nitride ($\text{SiO}_2/\text{SiN}_x$), silicon oxynitride/silicon nitride ($\text{SiO}_x\text{N}_y/\text{SiN}_x$) and aluminium oxide/silicon nitride ($\text{Al}_2\text{O}_3/\text{SiN}_x$) with various thicknesses are examined to produce optimise average reflectance (R_{avg}), short-circuit current density (J_{sc}) and efficiency (η). From the results, the device with 5 μm front upright pyramids, $\text{SiO}_2/\text{SiN}_x$ DLARC (60 nm/60 nm), and 1 μm rear upright pyramids exhibits R_{avg} , J_{sc} and η of 15.61%, 38.40 mA cm^{-2} and 21.02% respectively. The results demonstrate that the light-trapping strategies successfully enhance the efficiency of the thin TOPCon solar cells.

Keywords Silicon · Thin solar cells · TOPCon · SunSolve

1 Introduction

The International Technology Roadmap for Photovoltaics (ITRPV) reported that n-type silicon (Si) solar cells dominated the photovoltaic (PV) market in 2024, and their market share is expected to increase over the next decade [1]. For high-efficiency Si solar cells, the tunnel oxide passivated contact (TOPCon) cells gained a high reputation due to its high efficiency over the past few years with the standard 160 μm thickness of Si wafer [2]. TOPCon dominates the industry due to its capability to withstand high-temperature processes, compared to passivated emitter and rear cells (PERC) and heterojunction with intrinsic thin layers (HIT) counterparts [1, 3–6]. TOPCon, which includes ultra-thin

(< 2 nm) tunnel oxide (SiO_2) and phosphorus-doped polycrystalline-Si (poly-Si), increases the open-circuit voltage (V_{oc}), enhances passivation and electrical conductivity, hence increasing efficiency (η) [7]. The tunnel oxide also reduces the recombination current in the rear metal contact ($J_0 < 5 \text{ fA cm}^{-2}$) and suppresses contact resistivity ($\rho_c < 10 \text{ m}\Omega \text{ cm}^2$) [5].

The primary advantages of thin TOPCon cells share their cost-effectiveness and environmental friendliness, resulting from reduced material consumption. However, thin TOPCon cells suffer from poor light absorptance due to the thinner absorber layer, resulting in lower device efficiency. Light trapping strategies such as front surface texturing, anti-reflective coating (ARC) and rear surface texturing/rear reflector compensate for the efficiency loss due to the thinner Si absorber layer and poor light absorptance, making the efficiency of the thin TOPCon cells (< 150 μm) to be comparable with the conventional wafer thickness [8, 9]. Light coupling is vital for increasing absorptance and efficiency in the thin TOPCon cells by introducing micron- or sub-micron-scale structures on the device's surface, which was first inspired by the moth eye (protrusion array) to enhance

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sensitivity [10–12]. Since Si has appropriate hardness and reacts to chemical solutions (acid and alkaline), almost all kinds of manufacturing technologies can be used to fabricate surface texturing, including wet etching, dry etching, nano-imprinting and laser interference lithography [13–15]. The most famous geometrical shape used for surface texturing in industry is the upright pyramids, and it is not limited to other structures such as cylinders, cones, inverted pyramids, hemispheres, and three other advanced geometries, including mosaics, roses, and zigzags [16]. Apart from surface texturing, refractive index mismatch between the Si substrate and air can be mitigated by introducing an ARC [10]. Using the refractive index (n) matching, $n_1 = \sqrt{n_0 n_s}$, of air ($n = 1.0$) and Si ($n = 3.94$), quarter-wavelength theory, $nd = \frac{\lambda}{4}$ where d is thickness, the best ARC material for Si solar cells is silicon nitride (SiN_x) ($n = 1.99$) with 75 nm thickness. However, this method only reduces the reflectance within a narrow and selective wavelength. To achieve lower average reflectance (R_{avg}) across the 300–1200 nm spectrum, double layer anti-reflective coating (DLARC) is a superior method due to further reduction of abrupt changes in the refractive index to the Si substrate, which introduces more control points for interference and allows more complex destructive interference patterns that flatten the reflectance curve within the 300–1200 nm wavelength region [17]. Ultimately, in addition to using surface texturing to refract light at the front surface, the geometrical optics method also reflects light into the device from the rear of the cells. The rear reflector forms a specific angle with the front surface of the device, which reflects light to the front surface, thereby increasing the optical path length and enhancing photocurrent generation. This occurs when the incident angle exceeds the critical angle, resulting in total internal reflectance in the device [10].

The International Technology Roadmap for Photovoltaics (ITRPV) predicts that the wafer thickness of TOPCon cells will be reduced towards 110 μm by 2035, resulting in reduced material consumption and costs [1, 10, 18–21]. To date, the thinnest TOPCon cells recorded is 120 μm Si with an efficiency of 24%, using an M10 (182 mm) wafer, as reported by the Fraunhofer Institute for Solar Energy Systems (Fraunhofer ISE) in 2024 [22]. In 2025, Wang et al. fabricated 130 μm TOPCon cells that utilised laser-enhanced contact optimisation (LECO) to reduce contact resistance (R_c) between the front metallisation and Si wafer, with efficiency of 25.7% [23]. Another work on 130 μm Si wafer (M10) by Wang et al. utilised Jollywood's Special Injected Metallization (JSIM), a derivation of LECO, with an efficiency of 25.1% [24]. The highest recorded efficiency for TOPCon is 26.58% with large-area solar cells, surpassing those of PERC and HIT technologies [10, 25–27]. The TOPCon solar cells employed n-type industrial TOPCon (i-TOPCon) configuration, typically

featuring wafer thickness of around 140 μm [28]. However, there is no published work on the optical optimisation of thin TOPCon solar cells (with 100 μm thickness). The optimisation performed on the 100 μm -thin TOPCon cells in this work can be applied to current TOPCon technologies while advancing towards thinner wafers at an industrial scale, in line with the ITRPV trends.

In this work, SunSolve simulation platform is used to investigate the optical and electrical performance of 100 μm -thin TOPCon Si solar cells. Using SunSolve, the effects of upright pyramids heights for front surface texturing, DLARC with SiN_x underlayer, and upright pyramids heights for rear surface texturing on the optical and electrical performance are investigated within 300–1200 nm wavelength region. Optical performance, such as reflectance (R), absorptance (A), transmittance (T), and electrical performance such as external quantum efficiency (EQE), V_{oc} , short-circuit current density (J_{sc}), fill factor (FF) and η as well as current density–voltage (JV) are simulated. The optimum settings for all the three light trapping strategies combined are compared to the standard planar 160 μm and 100 μm thin TOPCon without any light trapping strategies (i.e. as a reference). The optimisations conducted on the 100 μm -thick TOPCon cells in this study can be incorporated into existing experimental TOPCon cells thinner than 150 μm to further enhance the efficiency.

2 Methodology

SunSolve is widely used among researchers and manufacturers within the PV community. The software extracts the optical and electrical performance of semiconductor materials and PV technologies. SunSolve solves optical simulations using Monte Carlo ray tracing for features comparable and larger than the wavelength of the incident light, such as surface texturing (1–10 μm). Individual light rays are traced from the front surface, refracted and reflected off textured surfaces (front and rear), and absorbed through different layers with decreasing intensities. The thin-film optics (Transfer Matrix Method) solves thin layers, such as ARC, silicon dioxide (SiO_2) passivation layers and poly-Si films, where wave interference effects dominate. It accurately calculates reflectance, transmittance and absorptance for planar multi-layer stacks. Both methods work simultaneously and solve the Fresnel equations (Eqs. 1, 2, 3 and 4) and the Beer-Lambert Law (5) as given below.

$$r_s = \frac{n_1 \cos \theta_i - n_2 \cos \theta_t}{n_1 \cos \theta_i + n_2 \cos \theta_t} \quad (1)$$

$$t_s = \frac{2n_1 \cos \theta_i}{n_1 \cos \theta_i + n_2 \cos \theta_t} \quad (2)$$

$$r_p = \frac{n_2 \cos \theta_i - n_1 \cos \theta_t}{n_2 \cos \theta_i + n_1 \cos \theta_t} \quad (3)$$

$$t_p = \frac{2n_1 \cos \theta_i}{n_2 \cos \theta_i + n_1 \cos \theta_t} \quad (4)$$

$$I(x) = I(0)e^{-\alpha x} \quad (5)$$

The derivation yields equations for the amplitude reflectance coefficient (r) and amplitude transmittance coefficient (t) for both s-polarisation and p-polarisation, where n_1 and n_2 are the refractive indices of medium 1 and medium 2, respectively, θ_i is the angle of incidence, and θ_t is the angle of transmittance. Beer-Lambert Law describes exponential decrease in light intensity with depth, where $I(x)$ is the light intensity at depth x , $I(0)$ is the light intensity at $x=0$, α is the absorption coefficient and x is the depth or optical path length.

As for electrical simulation, SunSolve integrates with other tools, such as PC1D, which solves the drift–diffusion equations (Eqs. 6 and 7) for charge carriers and Poisson's equation (Eq. 8) below, to calculate the internal electric field. It is also integrated with Quokka for modelling current flow and recombination in complex geometry, and Griddler, which focuses on electrical losses due to the metal grid. SunSolve also solves the current continuity equations (Eqs. 9 and 10) as shown below.

$$J_n = q\mu_c n \frac{\partial E_c}{\partial x} + qD_n \frac{\partial n}{\partial x} \quad (6)$$

$$J_p = q\mu_v p \frac{\partial E_v}{\partial x} - qD_p \frac{\partial p}{\partial x} \quad (7)$$

$$\frac{d}{dx} \epsilon_0 \epsilon_r = \frac{\partial y}{\partial x} = q(n - p) \quad (8)$$

$$\frac{\partial J_n}{\partial x} = q \left[R_n - G + \frac{\partial p}{\partial x} \right] \quad (9)$$

$$\frac{\partial J_p}{\partial x} = -q \left[R_p - G + \frac{\partial p}{\partial x} \right] \quad (10)$$

The electrostatic potential approximation, Poisson's equation, and the Shockley–Read–Hall (SRH) formalism are collectively employed in the model to simulate electrostatic behaviour, carrier recombination, and trap-assisted processes, enabling a detailed representation of trap state distributions and accurate device-level modelling [29].

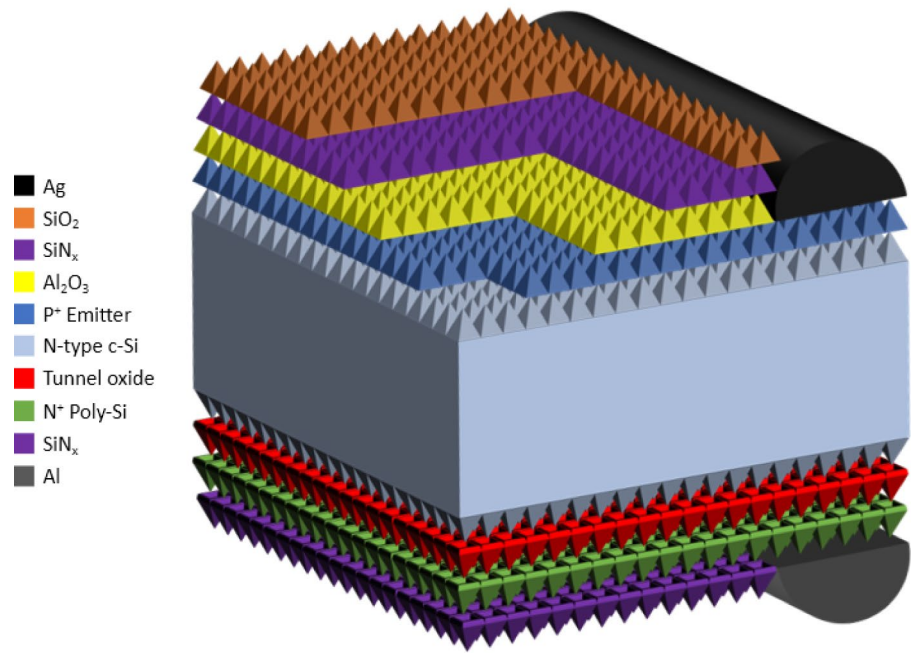
In this work, SunSolve platform is used to simulate thin TOPCon cells with monocrystalline silicon (c-Si) wafer thickness of 100 μm and a cell area (A_c) of 441 cm^2 . For the

simulation, the spectrum is set to sunlight of air mass 1.5G (AM1.5G) with 100 mW cm^{-2} illumination, and the operating temperature is set at 25 $^\circ\text{C}$ [29]. The incident angle is set to 0 $^\circ$, and the location is set to “Full area”. The device structure for TOPCon fixed parameters is extracted from Chen et al. and remains constant across the whole research [30]. The thickness of aluminium oxide (Al_2O_3) passivation layer, tunnel oxide, N^+ poly-Si and SiN_x dielectric layer is set to 8 nm, 1.5 nm, 100 nm and 90 nm, respectively [30]. The fixed layer thicknesses of the TOPCon cells are adopted and optimised following the methodology and results reported by Chen et al. The scattering parameters for the thin film and c-Si are set to fixed values with a fraction of 1, using the Lambertian distribution. The parameter at the optical interfaces of the electrodes is maintained with fixed scattering, incorporating optical shading fraction of 0.8 [31]. The simulation resolved the absorbance in each thin film layer, explicitly accounting for parasitic losses. Figure 1 (a) shows the configuration for a textured (front and rear) thin TOPCon solar cell with DLARC, while Fig. 1 (b) shows the configuration for a standard planar thin TOPCon solar cell with single-layer anti-reflective coatings (SLARC).

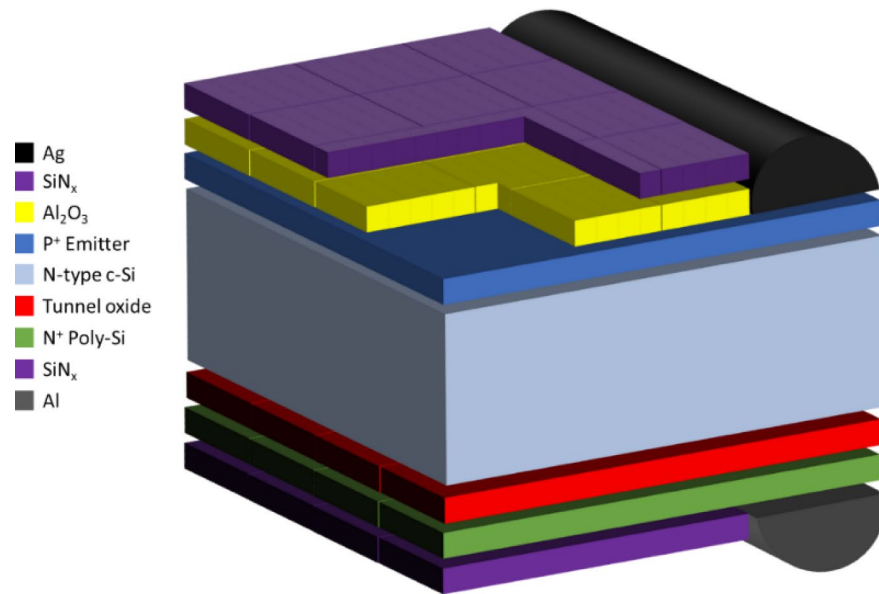
Regarding the electrical fixed parameters in the electrode and circuit tab within SunSolve, the contact layout for both the front and rear is set to “Grid”. The fingers and busbars parameters are set as in Table 1, and the series resistance and solar cells circuit are set as in Table 2.

Figure 2 shows the flow chart outlining the research process used to optimise the thin TOPCon cells in this work. For the first experiment, the front surface texturing geometry used is random upright pyramids (54.74 $^\circ$) with heights of 1 μm , 3 μm and 5 μm . The simulated TOPCon cells comprise a $\text{SiN}_x/\text{Al}_2\text{O}_3/\text{c-Si}/\text{SiO}_2/\text{poly-Si}/\text{SiN}_x$ stack from front to rear. The structure includes a 75 nm SiN_x ARC, a front silver (Ag) contact, and a rear aluminium (Al) contact. With the best results from the first experiment, top layer DLARC thicknesses of 20 nm, 40 nm and 60 nm are used, which are $\text{SiO}_2/\text{SiN}_x$ DLARC, silicon oxynitride 40% nitrogen (N)/ SiN_x ($\text{SiO}_x\text{N}_y/\text{SiN}_x$ DLARC) and $\text{Al}_2\text{O}_3/\text{SiN}_x$ DLARC, with the SiN_x underlayer set to 60 nm. The n and extinction coefficient (k) for all materials are sourced from the PV Lighthouse library. Table 3 provides the corresponding references for each material employed in the thin TOPCon cells simulated in this study. Finally, with the best DLARC, rear surface texturing of upright pyramids (54.74 $^\circ$) with heights of 1 μm , 3 μm and 5 μm is incorporated to the device. R curves, A curves, T curves, EQE curves, V_{oc} , J_{sc} , FF, and JV curves are simulated in the wavelength range of 300–1200 nm. With the R curves and JV characteristics, the R_{avg} and η are calculated respectively. All data points are simulated using 100,000 rays per simulation run, with results averaged to yield a representative value. The numerical precision is governed

Fig. 1 Configuration of **(a)** textured (front and rear) thin TOPCon solar cell with DLARC and **(b)** standard planar thin TOPCon solar cell with SLARC



(a)



(b)

by the number of rays used in the simulation, where higher numbers of rays leads to higher numerical precision (i.e. lower noise). Random texturing is implemented on both

the front and rear surfaces to emulate the surface morphology achieved in large-scale manufacturing via wet

Table 1 Parameters for fingers and busbars set in this simulation

Parameters	Fingers		Busbars	
	Front	Rear	Front	Rear
Number	128	168	12	12
Cross section	Rounded rectangular	Rounded rectangular	Rectangular	Rectangular
Height (μm)	15	15	15	15
Width (μm)	35	35	150	150
Finger pitch (cm)	0.1641	0.1250	NA	NA
Finger spacing (cm)	0.1606	0.1215	NA	NA
Pads	NA	NA	9	9
Pads length (μm)	NA	NA	1	1
Pad width (μm)	NA	NA	150	150

Table 2 Series resistance and solar cells circuit parameters set in this simulation

Parameters	Constant
Calculated series resistance, R_{sc} ($\Omega \text{ cm}^2$)	0.6932
Additional series resistance, R_{sa} ($\Omega \text{ cm}^2$)	0.3350
Total series resistance, R_s ($\Omega \text{ cm}^2$)	1.0282
Front skin sheet resistance ($\Omega \text{ sq}^{-1}$)	110
Rear skin sheet resistance ($\Omega \text{ sq}^{-1}$)	40
Base resistance, 100 μm ($\Omega \text{ cm}$)	2
Front electrode resistivity ($\Omega \text{ cm}$)	6.0×10^{-6}
Rear electrode resistivity ($\Omega \text{ cm}$)	2.6×10^{-6}
Light-collected current, J_L (mA cm^{-2})	38.40
Saturation current, J_{01} (pA cm^{-2})	0.08
Ideality factor 1, m_1 ($\Omega \text{ cm}^2$)	1
Shunt resistance ($\text{k}\Omega \text{ cm}^2$)	20

chemical texturing, consistent with the crystallographic orientation of the Si wafer.

3 Results and Discussion

Figure 3 (a), (b) and (c) show the R, A and T results for front surface texturing using the upright pyramids geometry with different heights of 1 μm , 3 μm and 5 μm . For the first parameter, the SiN_x thickness used is 75 nm, based on the quarter-wavelength theory at a wavelength of 600 nm [38]. Middle wavelengths are considered for the first parameter, as the visible light spectrum ranges from 400–800 nm. From Fig. 3 (a), at 600 nm wavelength, the lowest reflectance is achieved by 3 μm upright pyramids with an R value of 4.55%, followed by 1 μm and 5 μm with R values of 5.29% and 5.36%, respectively. As for R_{avg} on the wavelength

range of 300–1200 nm, 5 μm exhibits the lowest value of 18.69%. However, the difference in average reflectance from other heights is marginal. Compared to the planar front surface, the upright pyramids successfully reduce the R in the shorter wavelengths region ($< 600 \text{ nm}$), specifically in the blue, green, and ultraviolet (UV) regions. This is because the shorter wavelengths have a very high α value, as mentioned in the Beer-Lambert Law, where the light intensity drops rapidly with depth into the absorber layer [39, 40]. Hence, the upright pyramids front surface texturing is very important, especially in coupling shorter wavelengths light into the TOPCon cells to be absorbed by the c-Si on the first few hundred nanometres (surface) as shown in Fig. 3 (b), instead of being reflected away. The average absorptance (A_{avg}) is directly proportional to the upright pyramids' heights. As shown in Fig. 3 (c), the transmittance spikes in the longer wavelengths region. This is because c-Si has a bandgap of 1.12 eV at room temperature, which means only photons with energies greater than or equal to the c-Si bandgap will be absorbed [41]. This corresponds to wavelengths shorter than 1100 nm, and longer wavelengths will have insufficient energy to excite an electron across the bandgap, hence the higher transmittance [9, 42]. The difference in average transmittance (T_{avg}) for all the upright pyramids heights and planar front surfaces is very marginal, making surface texturing and ARC insignificant in reducing transmittance in the longer wavelengths region [43].

Figure 4 depicts the JV characteristics and Fig. 5 illustrates the EQE for front surface texturing using the upright pyramids geometry with heights of 1 μm , 3 μm and 5 μm respectively. From the reflectance curves, J_{sc} and EQE can be related to reflectance using the integral of the spectral generation rate and EQE equation, as shown in Eqs. 11 and 12 below.

$$J_{sc} = q \int_{\lambda_1}^{\lambda_2} EQE(\lambda) \cdot \phi_{\text{photon}}(\lambda) d\lambda \quad (11)$$

$$EQE(\lambda) = (1 - R(\lambda) - T_{\text{parasitic}}(\lambda)) \times IQE(\lambda) \quad (12)$$

where q is the elementary charge (coulomb), ϕ_{photon} is the incident photon flux of the solar spectrum, $T_{\text{parasitic}}(\lambda)$ is the transmittance through non-active layers or unabsorbed light through the active layer and IQE is the internal quantum efficiency. Higher reflectance results in lower EQE, assuming another factor is held constant, which also leads to lower J_{sc} . From the equation above, J_{sc} is directly proportional to the number of absorbed photons. With 5 μm upright pyramids, which demonstrates the lowest R_{avg} , the J_{sc} value is the highest (37.69 mA cm^{-2}) with the highest value of η (20.64%) when compared to 1 μm and 3 μm . Comparing the EQE to the absorptance results, the EQE is lower in the blue/UV and infrared (IR) regions. In the blue/UV

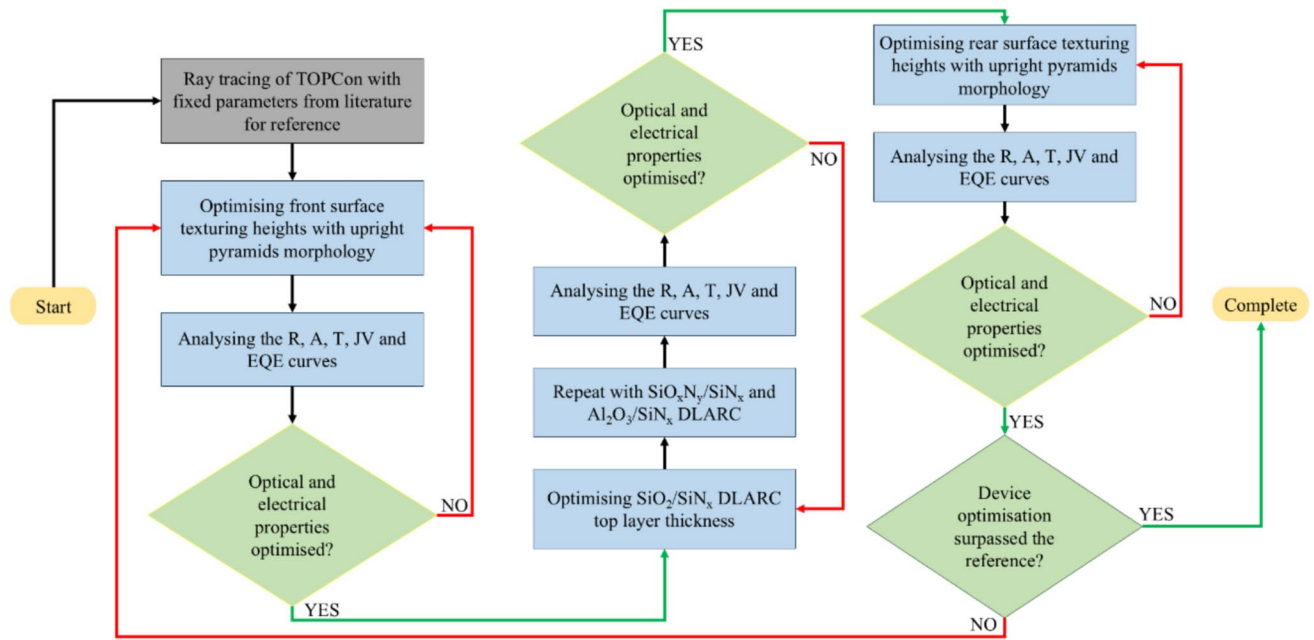


Fig. 2 Flow chart on optimisation of thin TOPCon cells

Table 3 Corresponding references for materials in the thin TOPCon cells

Material	Reference
Ag	Jia16 [32]
SiO ₂	Pal85e [33]
SiO _x N _y	SOPRA
Al ₂ O ₃	Kim97 [34]
SiN _x	Dut12 [35]
Al ₂ O ₃ (passivation)	Kum09 [36]
c-Si	Gree22 [37]
Poly-Si	Gre22 [37]
Al	Pal85b [33]

region, photons are absorbed close to the surface, making the response sensitive to front-surface recombination and parasitic absorptance. Poor surface passivation and significant parasitic absorptance reduce the EQE in the blue/UV region despite higher absorptance [2, 30]. In the IR region, photons penetrate deeply into the wafer and depend on long minority-carrier diffusion lengths and efficient rear-side collection [23, 44]. However, limited bulk lifetime, insufficient rear collection, or parasitic absorptance in the rear poly-Si and contact reduce the carrier collection probability and, consequently, the EQE at long wavelengths [23, 44]. Front surface texturing with 5 μm upright pyramids, when compared to lower heights, is more effective in reducing reflectance for shorter wavelengths. This is due to enhanced light absorptance since the 5 μm upright pyramids are much

larger than the wavelengths of the incident light [45]. This is explained by geometrical optics, where multiple reflectance occurs when the incident lights strike the pyramid facets and undergo several internal reflectance before eventually being coupled into the TOPCon cells and for increased absorptance [46]. A decrease in upright pyramids heights (same order of magnitude as the wavelength of the incident light) results in light behaving like a wave, where diffraction effects become more dominant and increase the reflectance [46]. Experimentally, it is feasible to fabricate upright pyramids of approximately 5 μm in height on 100 μm -thin TOPCon cells. Terheiden et al. demonstrated 100 μm Si solar cells (including ~ 5 μm pyramids), achieving an efficiency of 20% [47]. Apart from that, Woo et al. successfully incorporated pyramid textures ranging from 3 to 10 μm on 50 μm -thick wafers [48].

Figure 6 (a), (b), (c) show the R, A and T results for SiO₂/SiN_x, SiO_xN_y/SiN_x and Al₂O₃/SiN_x DLARC replacing the 75 nm SiN_x ARC, with 5 μm upright pyramids (from first parameter). The thickness used for SiN_x is reduced to 60 nm, as quarter-wavelength theory does not apply in DLARC, since the primary purpose is to minimise reflectance in the broadband range, rather than at a single wavelength. From Fig. 6 (a), all three DLARC reflectance curves with a 20 nm top layer are too thin to have an effect as optimised DLARC [49]. The reflectance is significantly higher than that of the SLARC in both the short wavelengths range (< 325 nm) and the longer wavelengths range (> 600 nm) until the IR range. The 40 nm thick top layer exhibits a mid-effect, and a 60 nm thick layer yields the most promising results in

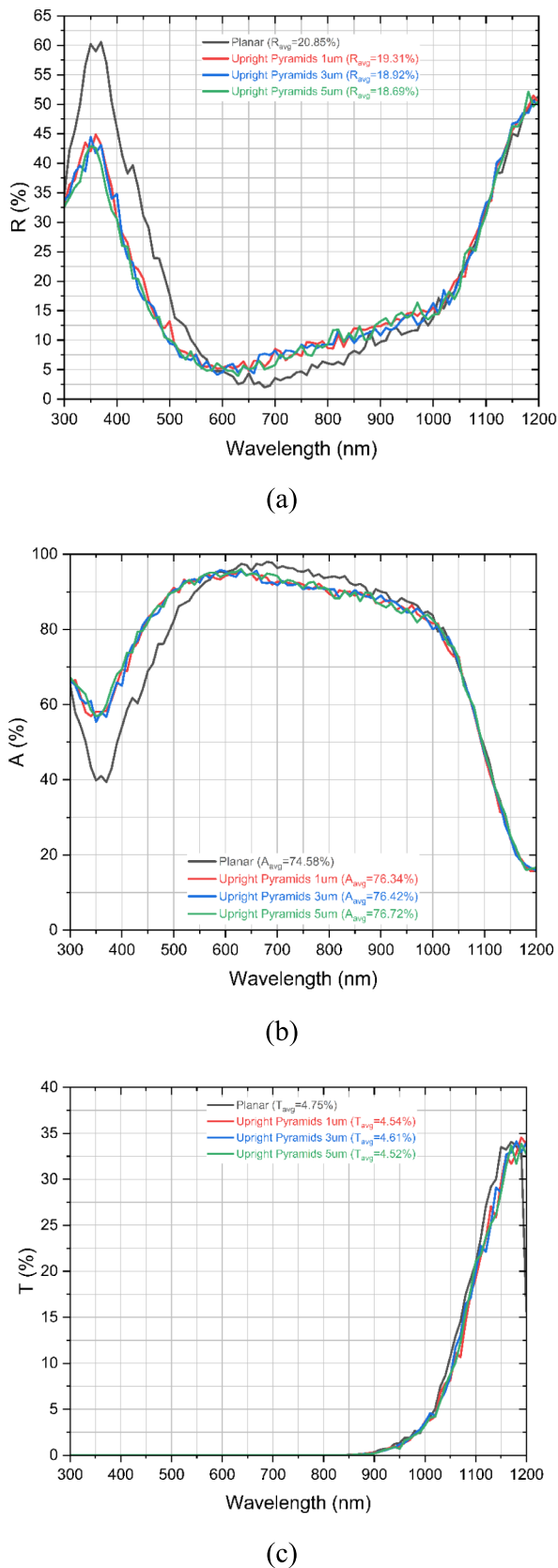


Fig. 3 (a) Reflectance (b) absorptance and (c) transmittance for upright pyramids front surface texturing with 1 μm , 3 μm , and 5 μm heights

reducing reflectance, with R_{avg} values of 16.18% ($\text{SiO}_2/\text{SiN}_x$), 15.99% ($\text{SiO}_x\text{N}_y/\text{SiN}_x$), and 15.58% ($\text{Al}_2\text{O}_3/\text{SiN}_x$). This also indicates that a less abrupt change in the refractive index of a thin film yields better DLARC materials. From the SunSolve library, the refractive index for the thin films at 600 nm is 1.46 for SiO_2 and 1.62 for SiO_xN_y and Al_2O_3 [33, 34]. As for the absorptance shown in Fig. 6 (b), the average performance for a 20 nm thick top layer is the lowest among the other thicknesses. The absorptance of $\text{SiO}_x\text{N}_y/\text{SiN}_x$ and $\text{Al}_2\text{O}_3/\text{SiN}_x$ at 40 nm and 60 nm is slightly lower than the SLARC in the mid-visible light spectrum, but then increases at longer wavelengths until the IR range. Meanwhile, for $\text{SiO}_2/\text{SiN}_x$, the absorptance is slightly higher compared to the other two DLARC in the mid-visible light spectrum, but barely higher than the SLARC at the longer wavelengths region, especially for 40 nm thickness. All three DLARCs with a 60 nm top layer depict an improvement in absorptance from 650 to 1050 nm, where the difference is marginal between each other. As for T_{avg} shown in Fig. 6 (c), the behaviour is the same as that of the previous parameter, which is 0% on most shorter wavelengths and increases in the near-infrared (NIR) and IR regions. The thickness of the top layer thin films for $\text{SiO}_2/\text{SiN}_x$, $\text{SiO}_x\text{N}_y/\text{SiN}_x$, and $\text{Al}_2\text{O}_3/\text{SiN}_x$ is inversely proportional to the R_{avg} and directly proportional to the A_{avg} [50]. So far, the TOPCon cells with 5 μm upright pyramids and $\text{Al}_2\text{O}_3/\text{SiN}_x$ (60 nm/60 nm) DLARC demonstrates the best optical performance.

Figure 7 illustrates the JV characteristics and Fig. 8 shows the EQE for $\text{SiO}_2/\text{SiN}_x$, $\text{SiO}_x\text{N}_y/\text{SiN}_x$ and $\text{Al}_2\text{O}_3/\text{SiN}_x$ DLARC with 20 nm, 40 nm and 60 nm top layer thicknesses. For 20 nm, the J_{sc} and η , $\text{SiO}_2/\text{SiN}_x$ is slightly underperformed, while $\text{SiO}_x\text{N}_y/\text{SiN}_x$ is on par with the SLARC. As for 20 nm $\text{Al}_2\text{O}_3/\text{SiN}_x$, the J_{sc} and η perform better, but only with a slight improvement. Given the optical performance of a 20 nm thickness for the top layer, it is insufficient to compensate for the reduced efficiency of the thin TOPCon cells. The J_{sc} for all three DLARC reached a peak value at a 60 nm top layer thickness, with the $\text{SiO}_2/\text{SiN}_x$ DLARC (60 nm/60 nm) outperforming the other two DLARCs. Although the $\text{SiO}_2/\text{SiN}_x$ DLARC exhibits higher reflectance and lower absorptance (by a small margin), η is the highest, with a value of 21.00%, due to its significantly higher J_{sc} value. The EQE curve exhibits behaviour similar to that observed in the previous experiment, showing lower values in the blue/UV and near-IR regions when compared with the absorptance curve. However, the 60 nm/60 nm DLARC for all material combinations are able to trap light more effectively, as indicated by the higher EQE values compared with the SLARC.

Table 4 below presents the JV characteristics and R_{avg} for $\text{SiO}_2/\text{SiN}_x$, $\text{SiO}_x\text{N}_y/\text{SiN}_x$ and $\text{Al}_2\text{O}_3/\text{SiN}_x$ DLARC with 60 nm thickness on top of 60 nm SiN_x . The V_{oc} value is the same, while FF only shows a slight difference between the

Fig. 4 JV characteristics for upright pyramids front surface texturing with 1 μm , 3 μm , and 5 μm heights

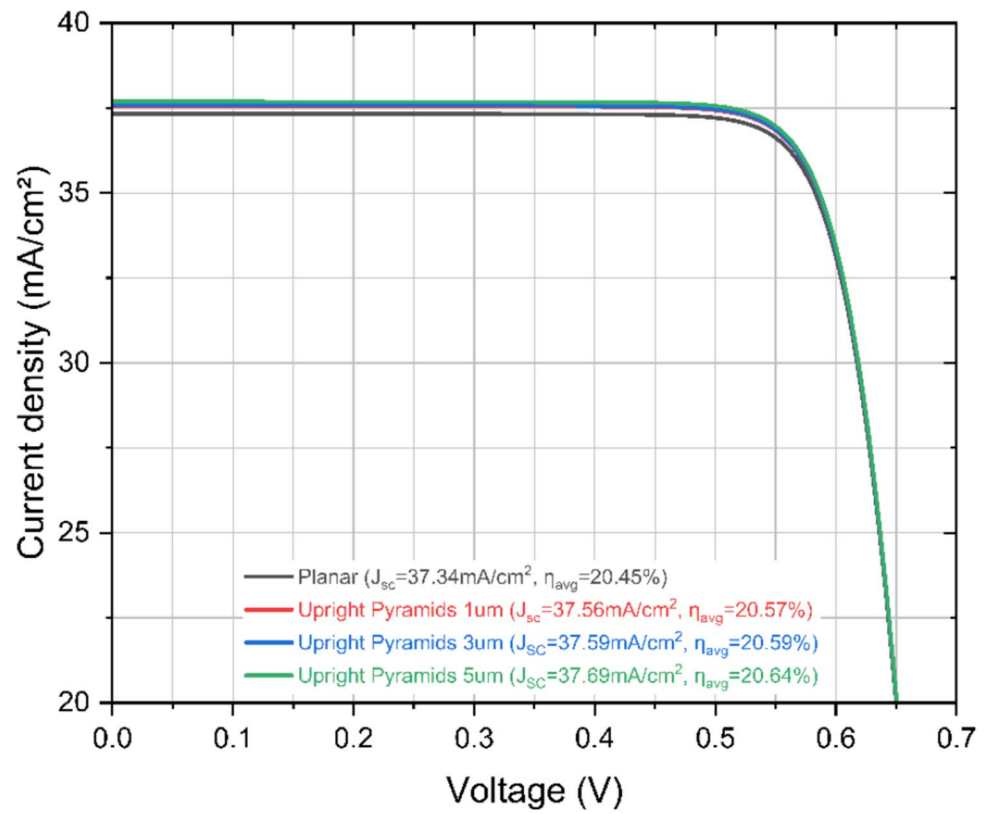
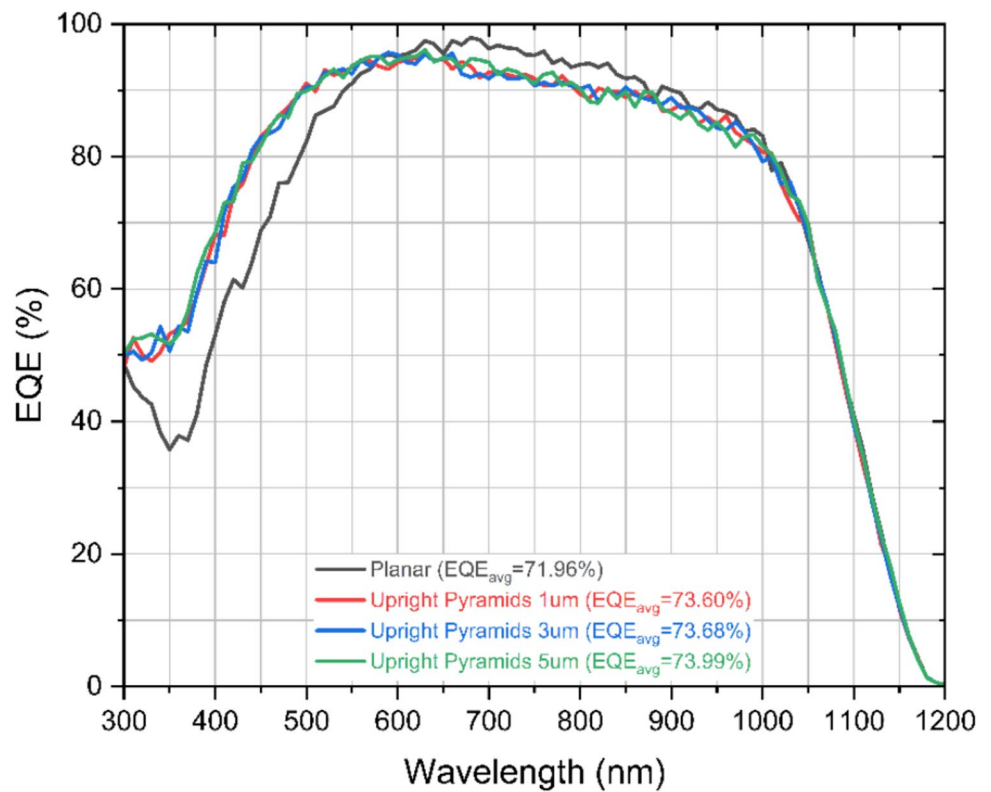


Fig. 5 EQE for upright pyramids front surface texturing with 1 μm , 3 μm and 5 μm heights



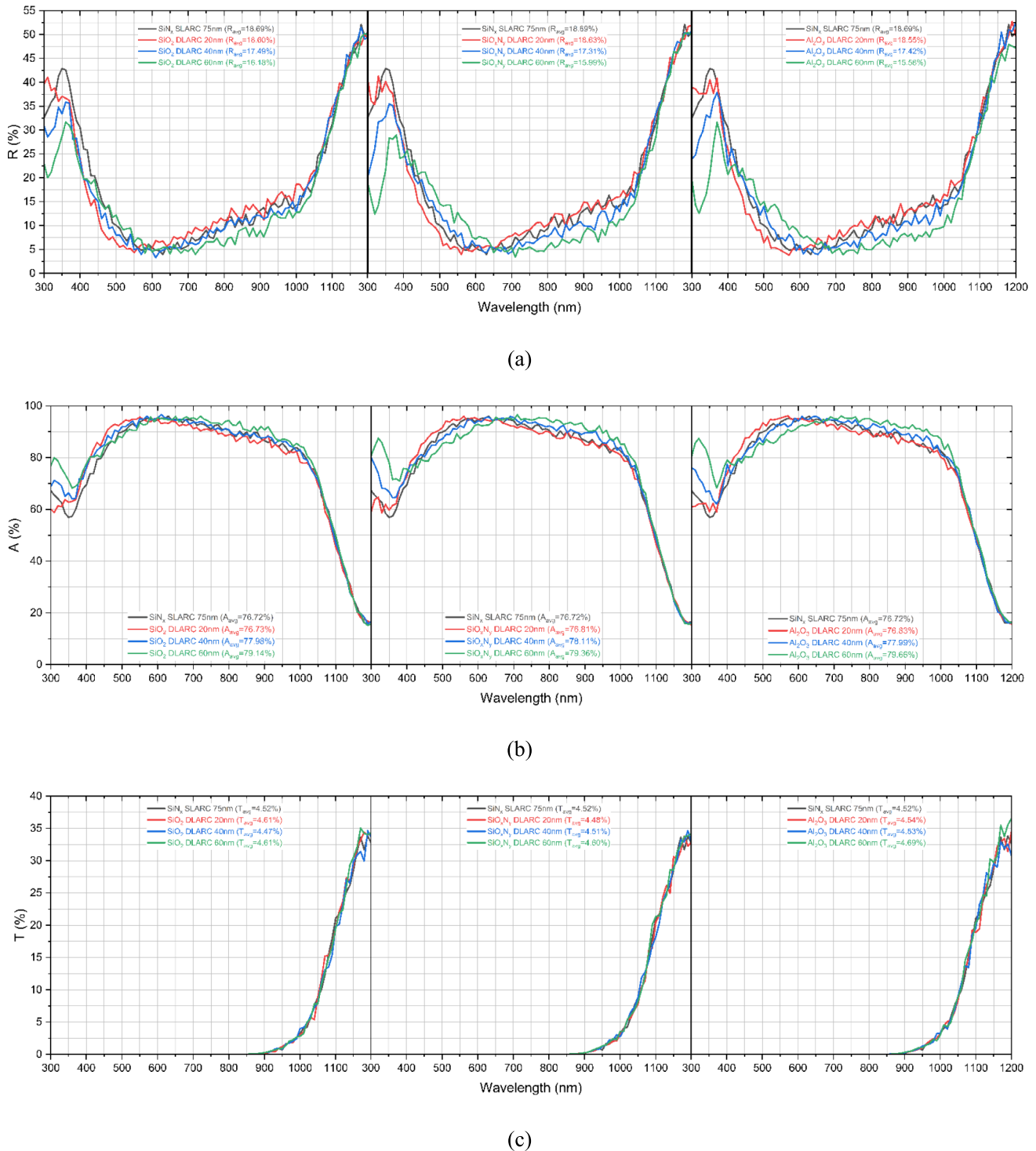


Fig. 6 (a) Reflectance (b) absorptance and (c) transmittance for DLARC ($\text{SiO}_2/\text{SiN}_x$, $\text{SiO}_x\text{N}_y/\text{SiN}_x$ and $\text{Al}_2\text{O}_3/\text{SiN}_x$), with the top layer thicknesses of 20 nm, 40 nm and 60 nm, and the bottom layer thickness of 60 nm

three DLARC. The EQE for a 60 nm top layer thickness, in the shorter wavelength range, indicates that $\text{SiO}_2/\text{SiN}_x$ has a smaller peak in the region below 400 nm but consistently remains higher than SLARC until around the 475 nm

region. Meanwhile, for $\text{SiO}_x\text{N}_y/\text{SiN}_x$ and $\text{Al}_2\text{O}_3/\text{SiN}_x$, the peak is higher, then decreases to a level lower than SLARC at around the 425 nm region. Although the TOPCon cells with 5 μm upright pyramids and $\text{Al}_2\text{O}_3/\text{SiN}_x$ (60 nm/60 nm)

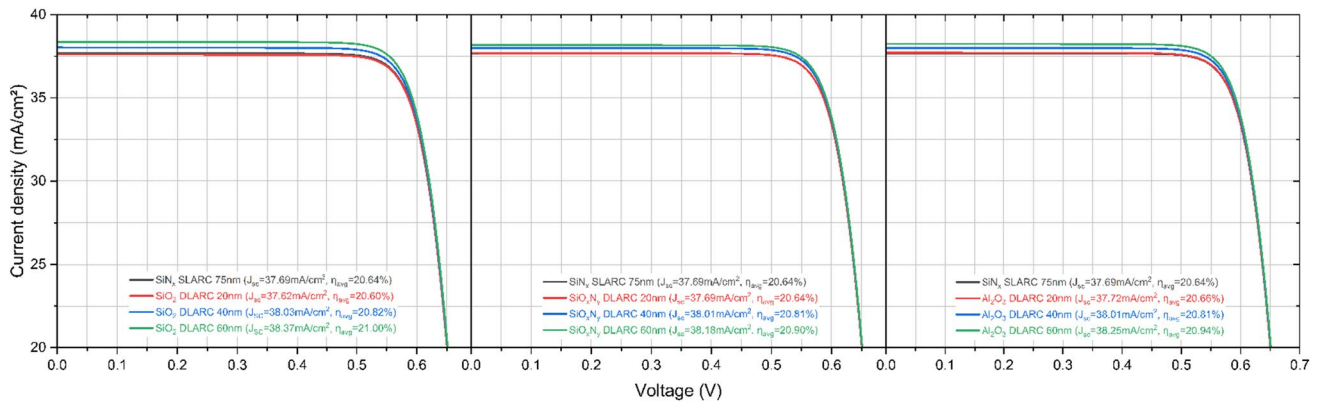


Fig. 7 JV characteristics for DLARC ($\text{SiO}_2/\text{SiN}_x$, $\text{SiO}_x\text{N}_y/\text{SiN}_x$ and $\text{Al}_2\text{O}_3/\text{SiN}_x$), with the top layer thicknesses of 20 nm, 40 nm and 60 nm, and the bottom layer thickness of 60 nm

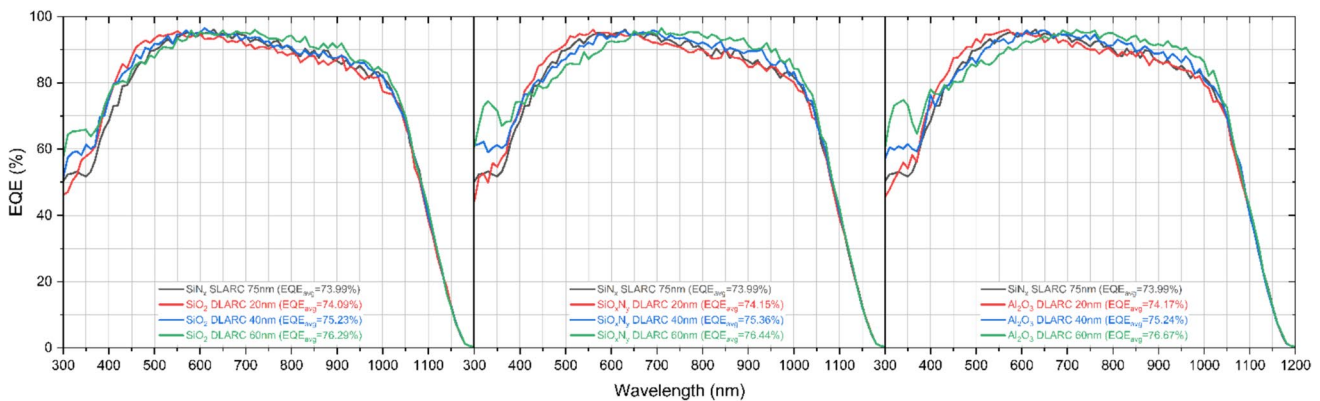


Fig. 8 EQE for DLARC ($\text{SiO}_2/\text{SiN}_x$, $\text{SiO}_x\text{N}_y/\text{SiN}_x$ and $\text{Al}_2\text{O}_3/\text{SiN}_x$), with the top layer thicknesses of 20 nm, 40 nm and 60 nm, and the bottom layer thickness of 60 nm

Table 4 JV characteristics and R_{avg} for DLARC for $\text{SiO}_2/\text{SiN}_x$, $\text{SiO}_x\text{N}_y/\text{SiN}_x$ and $\text{Al}_2\text{O}_3/\text{SiN}_x$ with 60 nm thickness on top of 60 nm SiN_x

Materials	η (%)	J_{SC} (mA/cm^2)	V_{OC} (mV)	FF (%)	R_{avg} (%)
SiN_x SLARC	20.64	37.69	691	79.31	18.69
SiO_2 DLARC	21.00	38.37	691	79.23	16.18
SiO_xN_y DLARC	20.90	38.18	691	79.25	15.99
Al_2O_3 DLARC	20.94	38.25	691	79.24	15.58

DLARC exhibit the best optical performance, the $\text{SiO}_2/\text{SiN}_x$ (60 nm/60 nm) DLARC demonstrates the best electrical performance and is considered for the third parameter. This is believed to be due to enhanced chemical passivation (lower interface defect density (D_{it})) and reduction in surface recombination velocity (SRV) at the p+emitter surface [51, 52]. The reduction in D_{it} and SRV improves the J_{sc} of the $\text{SiO}_2/\text{SiN}_x$ DLARC. Meanwhile, $\text{Al}_2\text{O}_3/\text{SiN}_x$ DLARC

introduces additional negative fixed charges and impede the hydrogen diffusion, leading to reduce passivation on the $\text{Al}_2\text{O}_3/\text{c-Si}$ interface [53, 54].

Figure 9 (a), (b), (c) depict the R, A and T results for rear surface texturing using the upright pyramids geometry with different heights of 1 μm , 3 μm and 5 μm . For the last parameter, the $\text{SiO}_2/\text{SiN}_x$ (60 nm/60 nm) DLARC, with 5 μm upright pyramids, is used. **Rear surface texturing is vital in reducing the reflectance in the longer wavelengths region** [43]. In Fig. 9 (a), the difference in reflectance reduction between the rear upright pyramids and the rear planar is increasing towards the 1200 nm wavelength region, with 5 μm having the lowest R_{avg} of 15.47%, followed by 1 μm (15.61%) and 3 μm (15.85%). Figure 9 (b) shows that the absorptance curves at longer wavelengths (> 1100 nm) for all rear upright pyramids with different heights are significantly higher compared to the planar rear counterparts, but there is little difference among them. Although 5 μm height has the lowest R_{avg} , 1 μm height

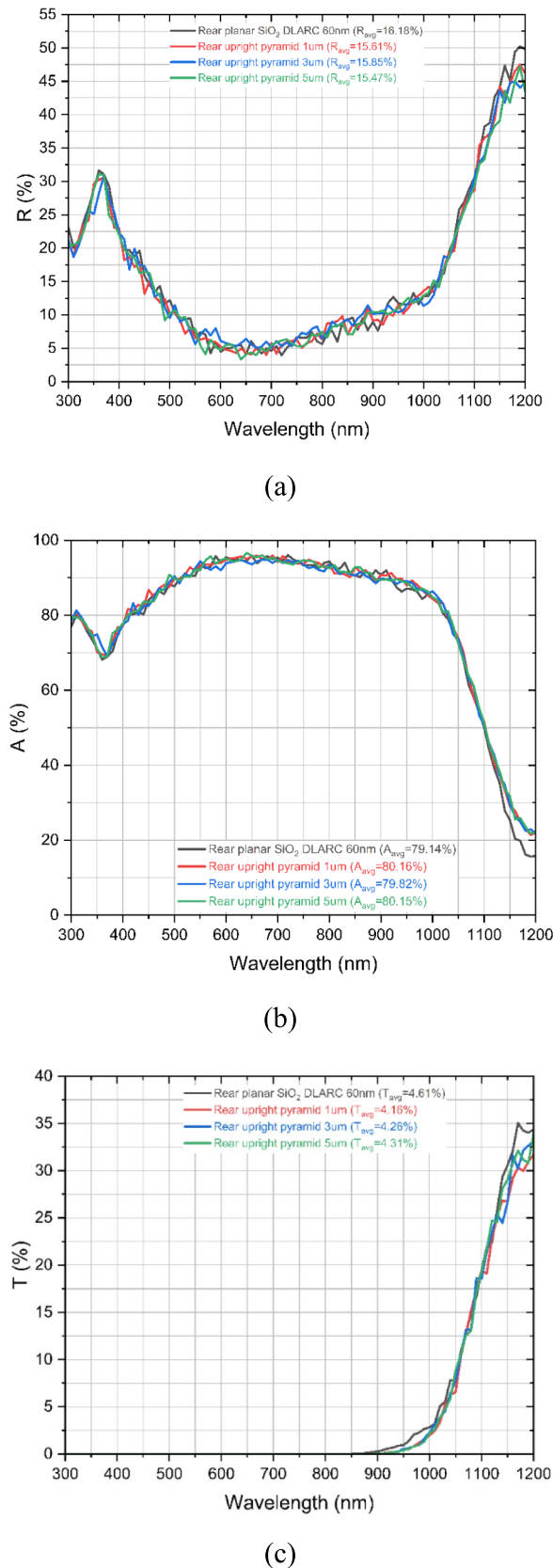


Fig. 9 (a) Reflectance (b) absorptance and (c) transmittance results for rear surface texturing (upright pyramids) with 1 μm , 3 μm , and 5 μm heights

outperforms in terms of A_{avg} by 0.01%. A similar case is observed in the transmittance curve in Fig. 9 (c), where 1 μm height has the least average transmittance (4.16%) due to light being reflected into the TOPCon cells from the rear, thereby increasing the light optical path length and allowing the absorber layer to absorb more light [44, 55]. In addition to the indirect bandgap and low absorption coefficient of Si, the thin Si layer in the TOPCon cells also weakly absorbs photons in the longer wavelengths region, allowing many photons to have a second pass through the device from the back [44]. Hence, rear surface texturing works by reflecting the light and increasing the light absorptance, especially in the longer wavelengths region, compared to previous parameters that focus on shorter wavelengths and broadband wavelengths [21, 44, 56].

Figure 10 shows the JV characteristics and Fig. 11 shows the EQE for rear surface texturing using the upright pyramids geometry with different heights of 1 μm , 3 μm and 5 μm . The highest J_{sc} and η are observed for the 1 μm rear upright pyramids height, which are 38.40 mA cm^{-2} and 21.02%, respectively. Then, the performance is followed by 5 μm height, with a 0.01 mA cm^{-2} and 0.01% marginal difference, and 3 μm as the least performing, with 0.21 mA cm^{-2} difference and 0.11% difference for η . Comparing the R_{avg} and η for both 1 μm and 5 μm rear upright pyramids, the 1 μm texture performs better despite having higher average reflectance than the 5 μm pyramids, due to improved passivation and lower recombination rate at the rear surface [57]. The 5 μm features possess a larger surface area with higher defect density at the tunnel oxide interface [58–63]. As for the EQE, the difference is also minimal, with the highest EQE_{avg} being 76.41% by 1 μm rear upright pyramids height. Like other parameters, the EQE curves are lower at shorter wavelengths and approach zero towards wavelength of 1200 nm due to the absorptance cut-off towards the bandgap of the c-Si.

Figure 12 shows the performance comparison for 100 μm thin TOPCon cells with optimised light trapping strategies from all the parameters, with non-optimised 100 μm and 160 μm standard TOPCon cells. The performance of the thin TOPCon cells surpassed that of the conventional 160 μm TOPCon cells, despite the absorber layer being reduced by 60 μm . The highest recorded η is achieved by 100 μm optimised TOPCon cells, with $21.02\% \pm 0.06\%$, with the lowest recorded R_{avg} of 15.61%. Therefore, the absolute efficiency gain of 0.29% obtained in the optimised 100 μm TOPCon cells is a meaningful enhancement, especially when considering its potential impact in the context of mass production of the solar cells [64]. The results demonstrate that the light-trapping strategies work efficiently by reducing optical losses within broadband wavelengths and by compensating for the efficiency loss in the thin TOPCon solar cells [65].

Fig. 10 JV characteristics for rear surface texturing (upright pyramids) with 1 μm , 3 μm , and 5 μm heights

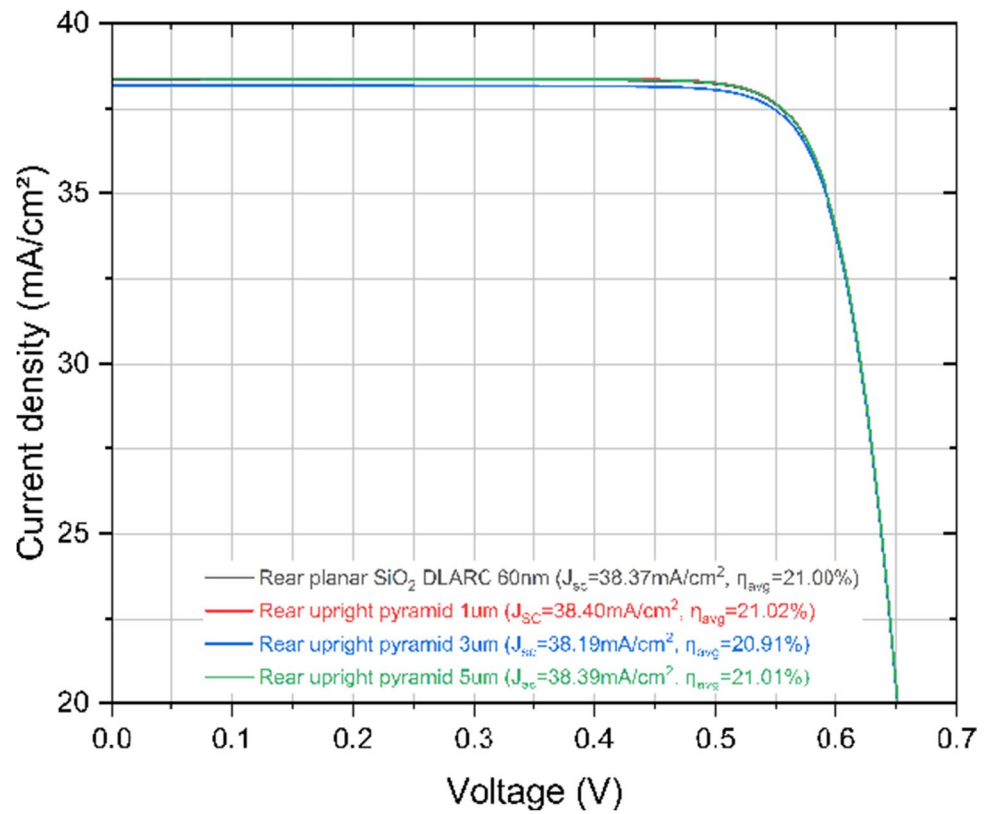
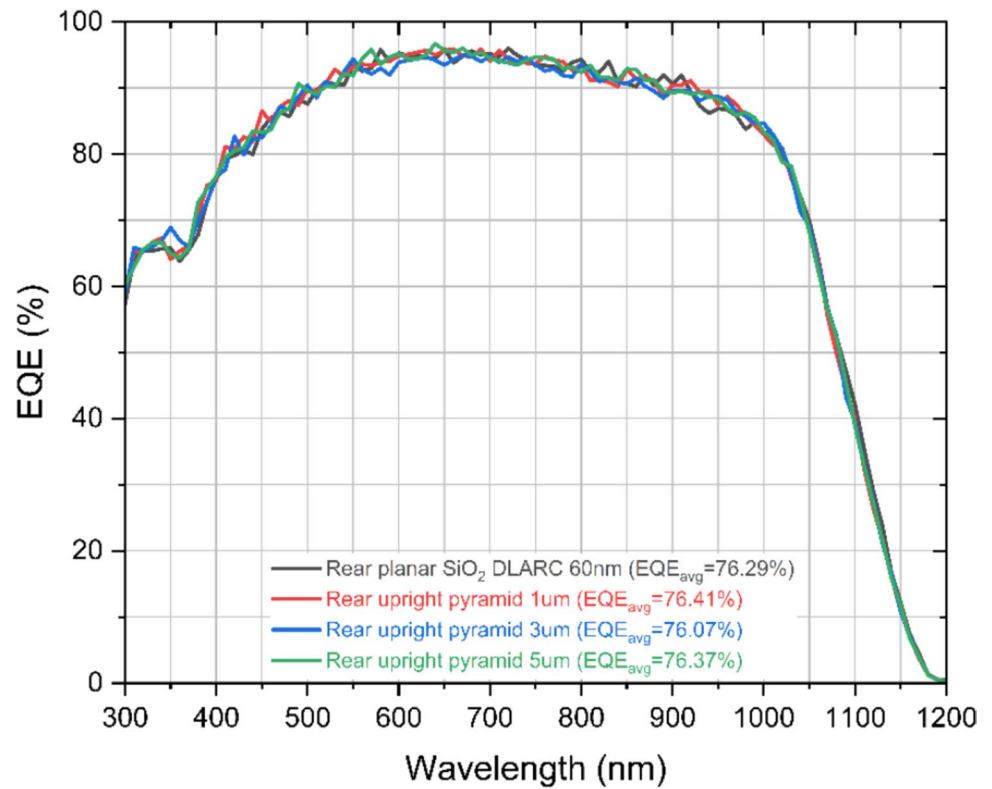


Fig. 11 EQE for rear surface texturing (upright pyramids) with 1 μm , 3 μm , and 5 μm heights



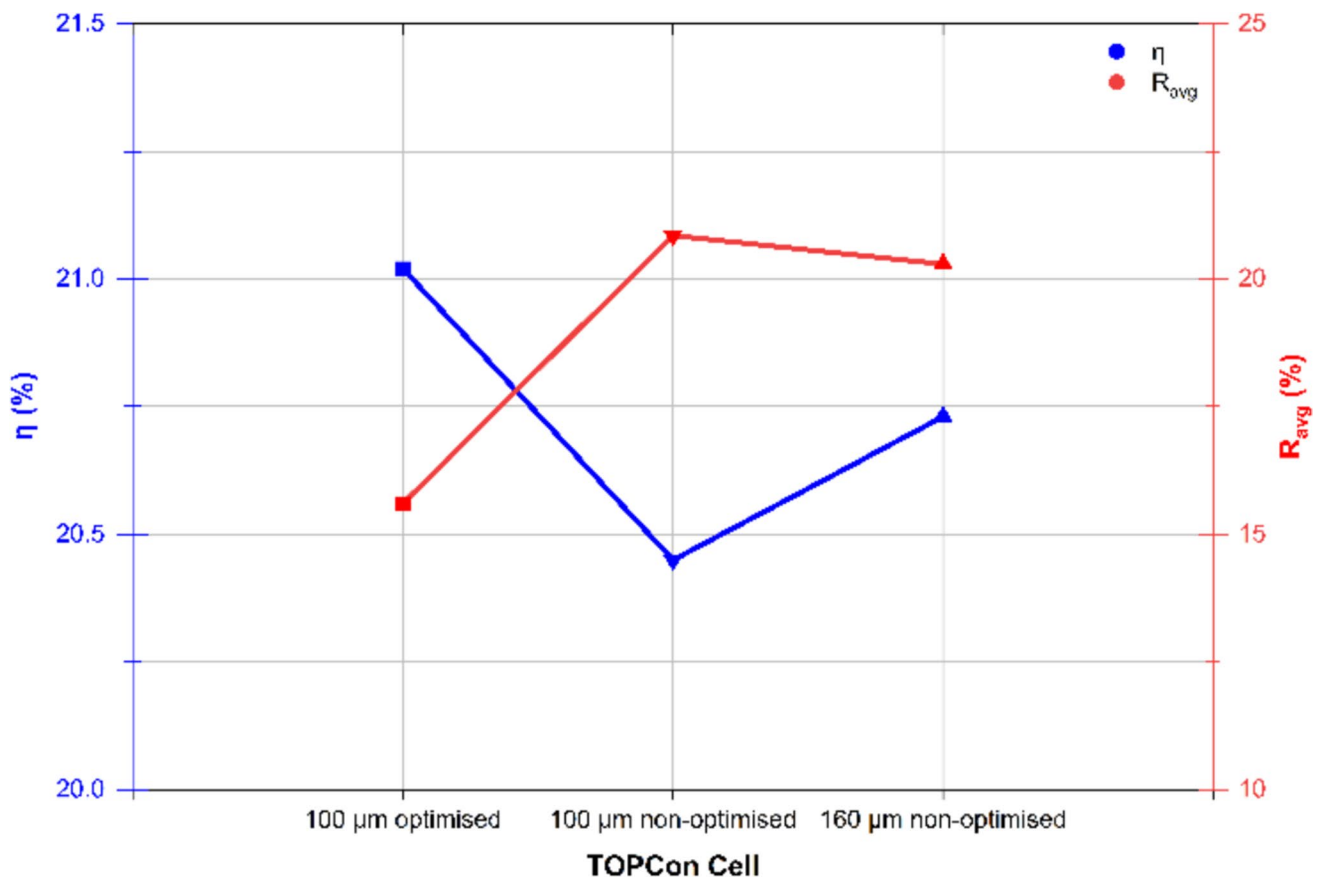


Fig. 12 Comparison of η and R_{avg} for 100 μm optimised light trapping strategies TOPCon cells with 100 μm non-optimised standard and 160 μm non-optimised standard TOPCon cells

4 Conclusion

This work aims to determine the optimal light-trapping strategies settings for 100 μm -thin TOPCon cells. Firstly, the front surface texturing with 5 μm upright pyramids is chosen because it reduces the R_{avg} to 18.69% and increases the η to 20.64%, surpassing the results achieved with 1 μm and 3 μm heights. The front surface texturing mitigates the optical loss in the shorter wavelength region when compared to the planar front surface. Then, with 5 μm upright pyramids on the front surface texturing, a $\text{SiO}_2/\text{SiN}_x$ (60 nm/60 nm) DLARC is incorporated to the device due to its highest η value of 21.00% when compared to $\text{SiO}_x\text{N}_y/\text{SiN}_x$ and $\text{Al}_2\text{O}_3/\text{SiN}_x$ DLARCs. Although $\text{Al}_2\text{O}_3/\text{SiN}_x$ (60 nm/60 nm) DLARC exhibits the lowest R_{avg} of 15.58%, $\text{SiO}_2/\text{SiN}_x$ (60 nm/60 nm) DLARC demonstrates higher J_{sc} of 38.37 mA cm^{-2} , hence the higher efficiency value. Introducing DLARC over SLARC reduces reflectance curves in the 300–1200 nm wavelength region for the thin TOPCon cells.

The overall performance of the thin TOPCon cells is increased by a small amount with the introduction of rear surface texturing with 1 μm upright pyramids. The upright

pyramids, with a height of 1 μm , yield the highest η value of 21.02% when compared to other heights, due to the improvements in optical performance in the longer wavelengths region. It has been observed that rear surface structuring increases the absorptance in the NIR and IR regions and decreases the transmittance by reflecting the light into the device, thereby increasing the optical path length for the light to be absorbed again by the absorber layer. The optimised 100 μm -thin TOPCon cells is equipped with 5 μm upright pyramids on the front surface texturing, $\text{SiO}_2/\text{SiN}_x$ (60 nm/60 nm) DLARC and 1 μm upright pyramids on the rear surface texturing, surpassing the performance of a 100 μm standard TOPCon cells and a standard conventional 160 μm TOPCon cells. The optimal light-trapping strategies developed in this work can be incorporated into the real 100 μm -thin TOPCon cells to enhance both optical and electrical performance.

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Data Availability Data will be made available by authors upon request.

Declarations

Competing Interests The authors declare no competing interests.

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